East West University

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Project Report Task -2

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Course Name : Machine Learning

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1.Executive summary

The notebook studies whether a small graph neural network (**STMG-Net**) can recognise six hand-grasp types from two-channel surface EMG (sEMG) *better than* classical machinelearning and deep-learning baselines, when the model is tested on a person it has never seen (Leave-One-Subject-Out, LOSO).

Cross-subject generalisation is the core difficulty. With no data from the new subject, every model reaches only ≈0.25–0.31 macro-F1 (chance 0.167), although the same models reach 0.89 accuracy under a random window split. That optimistic number is leakage-by-near-duplicates and is not reported as a result.

- **A little labelled data from the new subject changes everything.** Using 20 % of the new subject's trials as a calibration set (X_Cal) lifts the best model from 0.31 → 0.67 (simple up-weighting) → **0.866** (richer 64feature set + per-model tuning on X_Cal only).
- **The proposed STMG-Net did not beat the baselines in any setting** — zero-shot (0.244 vs 0.308), +Cal (0.390 vs 0.667) and +Adapt (0.702 vs 0.866). It lost on 0 of 5 held-out subjects each time, and the edge/graph structure gave no benefit in the ablation (0.244 with edges vs 0.254 without).
- **Leakage audit passed.** Subject, trial and window splits are disjoint; normalisation statistics for the test subject come from X_Cal only; permuted-label sanity checks fall at chance.

2. Data

The dataset is the public sEMG-for-basic-hand-movements data (Database 1): **5 healthy subjects** (female_1–3, male_1–2), each performing **6 grasps** — Cylindrical, Tip, Hook, Palmar, Spherical, Lateral — recorded on **2 channels at 500 Hz**. Each subject contributes 180 trials (30 per class) of 3 000 samples (6 s). The classes are perfectly balanced (max/min ratio = 1.0), there are no NaN/Inf values, but the signal amplitude differs strongly between people (e.g. channel-1 RMS from 0.29 to 0.82), which motivates persubject normalisation.

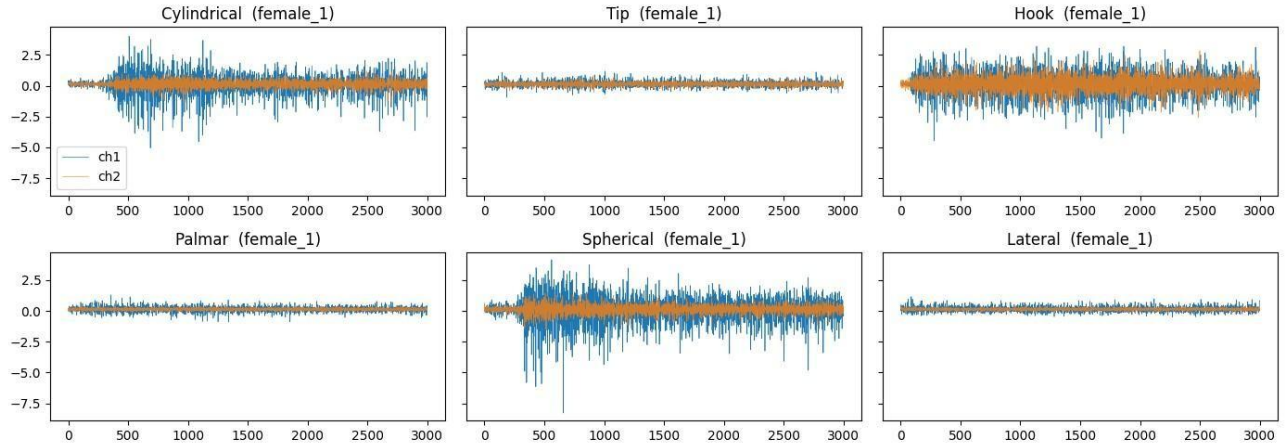


Figure 1 — One raw trial per grasp class (subject female_1), channels 1 and 2.

3. Methodology

3.1 Windowing and features

- **Windows:** 250 samples (0.5 s) with a hop of 125 samples (50 % overlap) → 23 windows per trial → **20 700 windows** in total; each window is mean-centred.
- **Base tabular features (22):** per channel the log-MAV, log-RMS, log-WL, log-VAR, zero crossings, slope-sign changes, Willison amplitude; spectral mean / median / peak frequency; plus inter-channel correlation and log-RMS ratio.
- **Extended features (64):** adds 8 log-band-power ratios (5–251 Hz), 4 AR coefficients, Hjorth mobility/complexity, skewness, kurtosis and peak amplitude per channel, and a band-power difference between channels. Used in the final "+Adapt" experiments.
- **Graph input for STMG-Net:** each window is split into 5 sub-segments per channel → **10 nodes**, each with 7 time-domain features; edges link neighbouring segments of one channel (temporal) and the same-time segments of the two channels (weighted by their absolute correlation).

3.2 Evaluation protocol

- **LOSO (5 folds):** train on 4 subjects (16 560 windows), test on the 5th. One training subject serves as validation for early stopping — never the test subject.
- **Calibration split:** of the held-out subject's trials, 20 % (6 per class = 36 trials, 828 windows) form **X_Cal**; the remaining 144 trials (3 312 windows) are **X_Test**. The split is by *trial*, never by window.
- **Per-subject Z-normalisation:** for the held-out subject the mean/std come from X_Cal only.
- **Three settings:** *Zero-shot* (no data from the new subject), *+Cal* (X_Cal added to training with upweighting), *+Adapt* (extended features; hyper-parameters, sample weights and fine-tuning epochs chosen by trial-wise CV *inside* X_Cal).
- **Metrics:** accuracy, balanced accuracy, macro precision/recall/F1, MCC, Cohen's κ, onevs-rest AUC, trainvs-test gap, run time. The headline metric is macro-F1 averaged over the 5 held-out subjects.

Models Used

In total 11 models were compared: 8 classical machine-learning models, 2 deep-learning baselines and the proposed graph network. Classical models were first run with default settings, then in a strongly regularised version.

Classical models

- Logistic Regression (LogReg): linear baseline; strongest default zero-shot model (macro-F1 0.304)
- SVM with RBF kernel (SVM-RBF)
- k-Nearest Neighbours (kNN): 101-NN when regularised; best model in the +Cal setting (0.667)
- Decision Tree: shallow trees when regularised
- Random Forest: slowest to fit in +Adapt (459 s)

- HistGradientBoosting (GradBoost)
- XGBoost
- LightGBM

Deep baselines

- MLP (Multi-Layer Perceptron): about 10k parameters (26k when fine-tuned); best final model (0.866)
- 1-D CNN: small CNN on raw windows, about 2.7k to 9.7k parameters

Proposed model

- STMG-Net (Spatio-Temporal Graph Network): 5,096 parameters

STMG-Net architecture: each window is split into 5 sub-segments per channel, giving 10 graph nodes with 7 time-domain features each. Edges connect neighbouring segments within a channel (temporal) and sametime segments across the two channels (weighted by absolute correlation). The network has a linear node encoder with a learned node embedding, 2 residual graph-convolution layers (LayerNorm, ReLU, dropout 0.3) with learnable temporal and cross-channel edge weights, a mean and max read-out, and a 2-layer classifier. Training used AdamW (learning rate 2e-3, weight decay 1e-3), label smoothing 0.05, classweighted cross-entropy and early stopping (patience 8). An ablation replaced the adjacency with the identity matrix ("no edges").

Fine-tuning (+Adapt): deep models (MLP, STMG-Net) were pre-trained on the other subjects and then fine-tuned on X_Cal (5x sample weight, learning rate 5e-4, 3,000 replayed source windows per epoch to avoid forgetting). Hyper-parameters and the number of fine-tuning epochs were selected by trial-wise crossvalidation inside X_Cal only.

4. Results

4.1 Stage 1 — default baselines, zero-shot

With default hyper-parameters the flexible models fit the training subjects well but collapse on a new subject: Random Forest reaches 0.857 train F1 but only 0.282 on the held-out subject (gap 0.575). Logistic Regression is the "baseline to beat" at macro-F1 **0.304**.

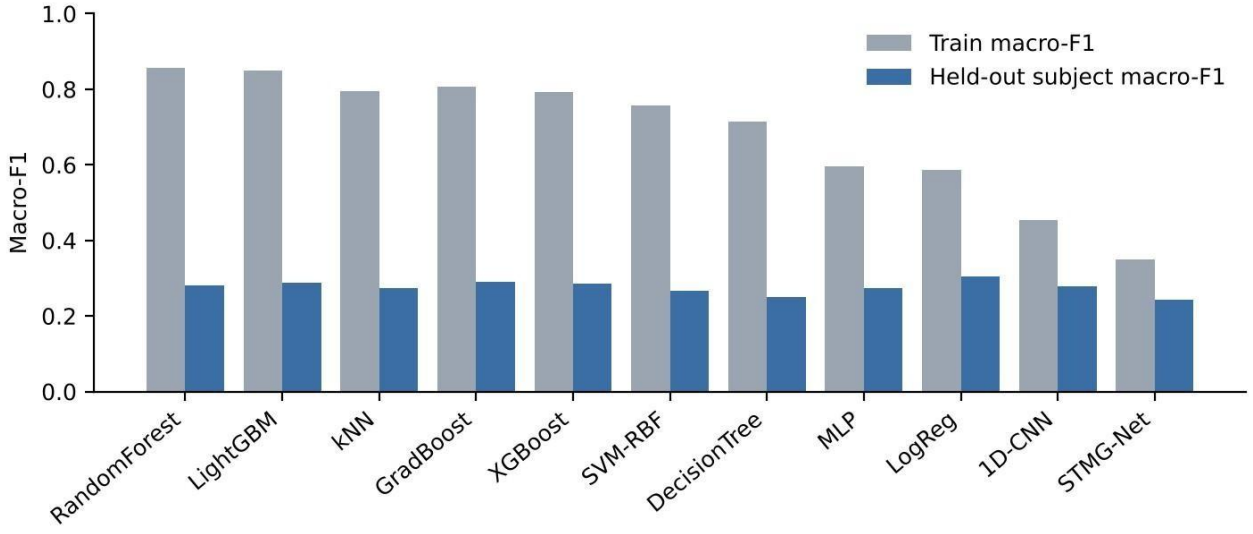


Figure 2 — Train vs. held-out-subject macro-F1 for default models (zero-shot). Large gaps indicate subjectspecific over-fitting.

Model	Acc	Macro-F1	MCC	AUC	Train F1	Gap
LogReg	**0.321**	**0.304**	**0.190**	**0.728**	**0.586**	**0.282**
GradBoost	0.297	0.290	0.158	0.696	0.807	0.517
LightGBM	0.297	0.288	0.158	0.695	0.849	0.560
XGBoost	0.291	0.286	0.151	0.697	0.792	0.507
RandomForest	0.288	0.282	0.147	0.693	0.857	0.575
1D-CNN	0.327	0.278	0.208	0.737	0.454	0.176
MLP	0.287	0.273	0.147	0.703	0.595	0.322
kNN	0.281	0.273	0.139	0.663	0.795	0.522
SVM-RBF	0.277	0.266	0.135	0.684	0.756	0.490
STMG-Net (no edges)	0.323	0.254	0.205	0.759	0.351	0.097
DecisionTree	0.253	0.251	0.105	0.671	0.714	0.463
STMG-Net	**0.306**	**0.244**	**0.181**	**0.742**	**0.349**	**0.106**

4.3 Stage 3 — regularised models (zero-shot)

Shrinking model capacity (shallow trees, heavy L2, 101-NN, dropout 0.5, weight decay) narrows the gap, but the ceiling stays near 0.31. The best regularised baseline is the MLP at **0.308**, followed by Random Forest 0.305, SVM 0.301 and Logistic Regression 0.300; STMG-Net (0.244) is 0.065 behind and below the best baseline on all 5 subjects (female_3 by −0.211). No model is meaningfully subject-independent with only 2 sEMG channels and 4 training subjects.

4.4 Stage 4 — calibration (+Cal)

Adding the labelled X_Cal windows (5× weight) to the training set produces a large jump. kNN +Cal is best at **0.667** macro-F1 (AUC 0.911), followed by LightGBM 0.616, GradBoost 0.595 and Random Forest 0.594.

STMG-Net +Cal reaches 0.390 and is last, 0.276 below kNN; it loses on every held-out subject (−0.138 to −0.380).

Model (+Cal)	Macro-F1	Acc	MCC	AUC
kNN	**0.667**	**0.668**	**0.605**	**0.911**
LightGBM	0.616	0.622	0.550	0.892
GradBoost	0.595	0.605	0.530	0.885
RandomForest	0.594	0.605	0.532	0.886
XGBoost	0.575	0.587	0.510	0.876
LogReg	0.543	0.556	0.474	0.864
MLP (reg)	0.520	0.532	0.447	0.863
SVM-RBF	0.515	0.544	0.467	0.851
1D-CNN (reg)	0.487	0.508	0.430	0.879
DecisionTree	0.479	0.500	0.410	0.834
STMG-Net	**0.390**	**0.437**	**0.345**	**0.827**

4.5 Stage 5 — full adaptation (+Adapt), the final comparison

With the 64-feature set and every model tuned or fine-tuned using *only* trial-wise crossvalidation inside X_Cal, performance rises to the 0.83–0.87 range. Four model families are statistically indistinguishable at the top (MLP 0.866, LightGBM 0.863, XGBoost 0.862, GradBoost 0.860, kNN 0.858 — all within 0.008). STMG-Net +Adapt scores **0.702**, the lowest of 11 models, 0.165 below MLP, and is worse on every held-out subject (−0.114 to −0.300), with the highest variance (sd 0.132; female_2 only 0.491).

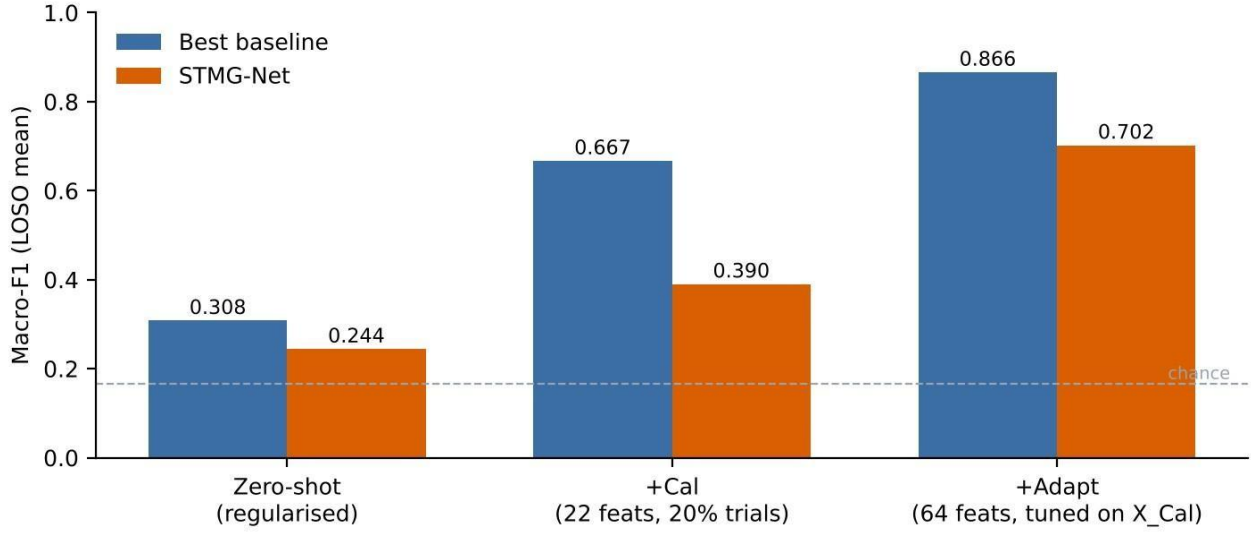
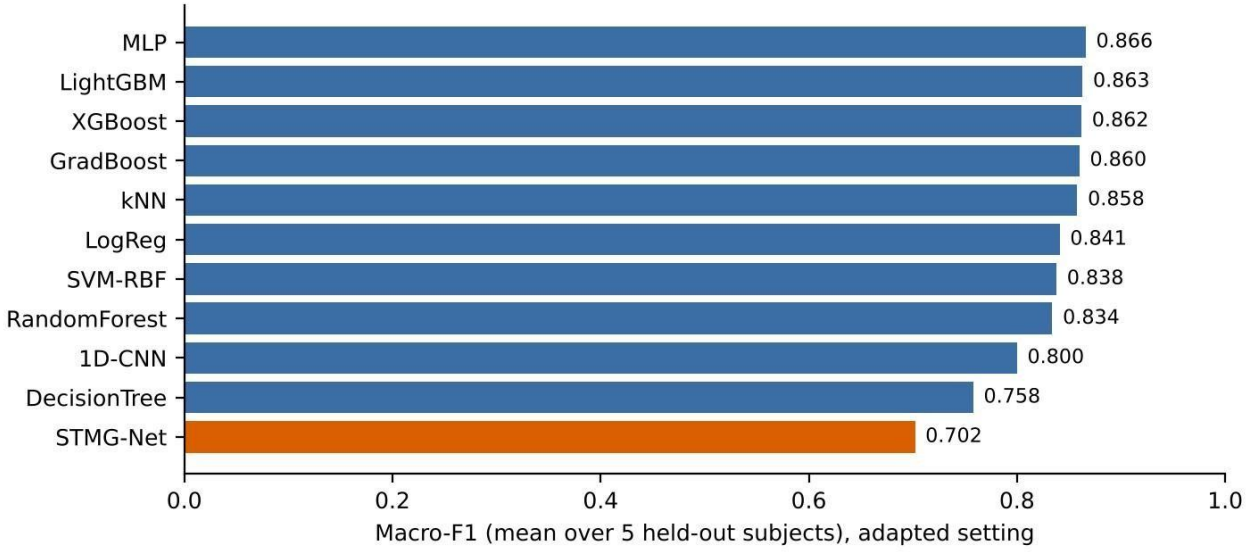
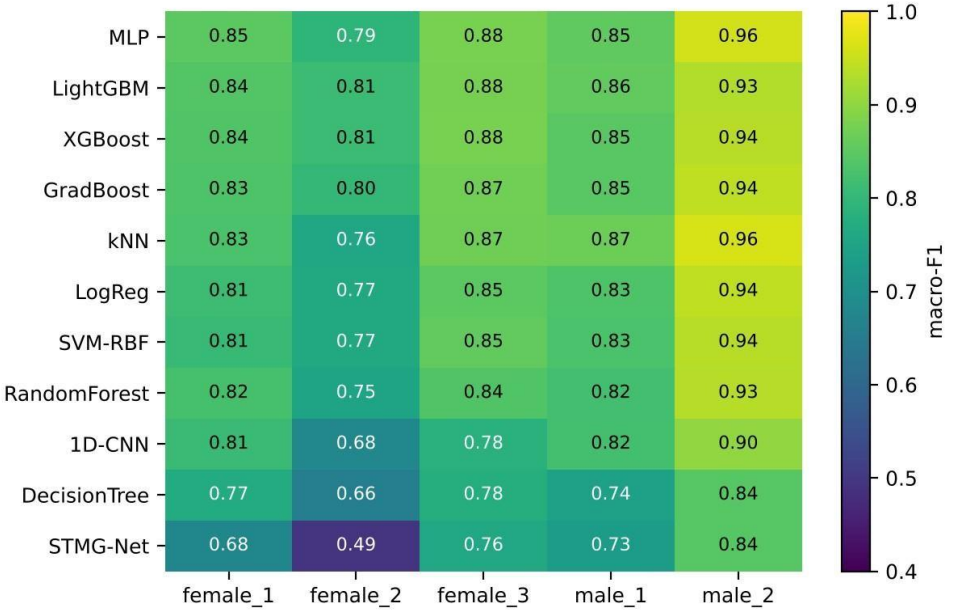


Figure 4 — Best baseline vs. STMG-Net at each stage (gap: 0.065 zero-shot, 0.276 +Cal, 0.165 +Adapt). STMGNet never overtakes.



FModel (+Adapt)	Acc	Macro-F1	MCC	AUC	Train F1	Fit time (s)
MLP	**0.867**	**0.866**	**0.842**	**0.982**	**0.746**	**4.9**
LightGBM	0.864	0.863	0.838	0.980	0.877	15.1
XGBoost	0.863	0.862	0.837	0.979	0.797	22.1
GradBoost	0.860	0.860	0.834	0.979	0.873	31.7
kNN	0.859	0.858	0.832	0.969	0.829	2.2
LogReg	0.841	0.841	0.811	0.976	0.759	41.9
SVM-RBF	0.839	0.838	0.808	0.961	0.938	61.8
RandomForest	0.834	0.834	0.803	0.975	0.845	459.3
1D-CNN	0.805	0.800	0.769	0.960	0.449	7.2
DecisionTree	0.758	0.758	0.712	0.924	0.587	11.2
STMG-Net	**0.713**	**0.702**	**0.667**	**0.938**	**0.395**	**9.5**



5. Integrity Checks

An automatic check was run at the end of the notebook, and all checks passed:

- No subject or trial appears in more than one of train, calibration and test sets, in all 5 folds.
- The tuning and early-stopping data come only from X_Cal and are split by trial.
- The test subject's normalisation uses X_Cal only. As a test, we added +1000 to X_Test and no train or X_Cal value changed.
- The scaler and imputer were fitted on training subjects only.
- Permuted-label test gave chance-level results (LogReg 0.184, kNN 0.185, RF 0.165, chance 0.167).

6. Why Did STMG-Net Perform Worse?

- **Too little input information.** Each node has only 7 simple features from a very short 50-sample piece of signal. The other models use 64 features (including spectral and band-power features) from the whole window. STMG-Net was not given these extra features, so the comparison favours the baselines.
- **The graph is too small.** It has only 10 nodes in two short chains, so there is not much for the graph layers to learn from. The model without edges scored the same, which is the clearest proof.
- **Under-fitting.** STMG-Net's training F1 was only 0.35 to 0.45, while other models reached 0.75 to 0.94. With just 5,096 parameters, dropout and strong weight decay, it could not even fit the training subjects well.
- **People differ too much.** With only 4 training subjects and 2 channels, the difference between people is bigger than the difference between grasps. Only data from the new person (X_Cal) fixes this, and kNN, boosted trees and the fine-tuned MLP use that data better than the graph network.

7. Limitations

- Only 5 subjects from one dataset. Differences of 0.01 or less between top models are smaller than the normal variation between subjects (sd 0.045 to 0.074), so these models should not be ranked against each other. The results are indicative, not statistically conclusive.
- The +Cal and +Adapt results need 36 labelled trials (6 per grasp) from every new user. This is a real cost, and zero-shot accuracy is still poor.
- STMG-Net was not tuned as much as the 64-feature baselines. So the result shows that this particular configuration performs poorly, not that graph models cannot work on sEMG.

8. Conclusions and Recommendations

- Subject-independent grasp recognition with 2 sEMG channels is not solved. Zero-shot macro-F1 stays around 0.3 for all 11 models, so only LOSO results should be reported.
- A short calibration for each user is the key: macro-F1 goes from 0.31 to 0.87. The best final model is the fine-tuned MLP (macro-F1 0.866, MCC 0.842, AUC 0.982, about 5 s per subject). LightGBM, XGBoost and kNN perform about the same.
- The proposed STMG-Net did not beat any baseline. This is a negative result, supported by the edge ablation and a leakage-free protocol.
- **Future work:**
    - Give STMG-Net the extended spectral and AR features at each node.
    - Use richer graphs (more segments, frequency-band nodes, learned adjacency).
    - Increase model size (hidden size 64 to 128, 3 layers).
    - Use data augmentation, domain-adversarial training or test-time normalisation.
    - Tune it with the same inner cross-validation on X_Cal as the baselines.

    - Add more subjects, or use the multi-day Database 2